

Project Design Phase

Solution Architecture

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Project Name: Plugging into the Future: An Exploration of Electricity Consumption Patterns

Solution Architecture Overview

This solution architecture illustrates the end-to-end flow of electricity consumption data. The dataset is collected from local storage or cloud storage, processed using Tableau, and transformed into interactive dashboards. The dashboards are embedded into a Flask-based web application, allowing end users such as energy analysts, researchers, and policymakers to explore electricity consumption patterns through maps, charts, and filters.

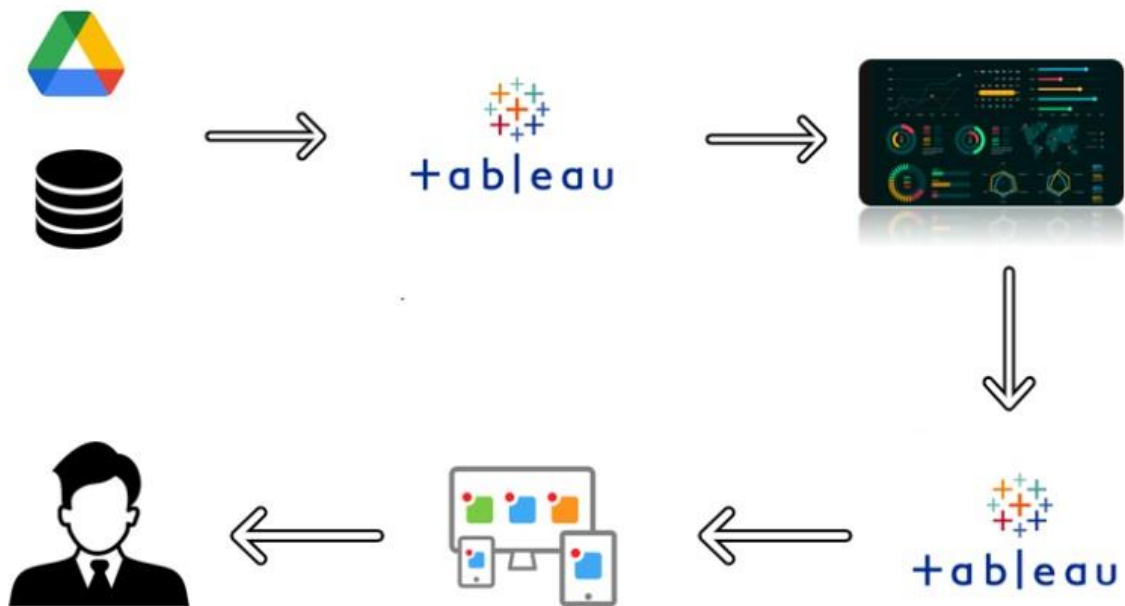


Figure 1: Solution Architecture for the Electricity Consumption Analysis System

Architecture Components

Component	Description
Data Source	Electricity consumption data stored in CSV files or cloud storage.
Data Processing	Tableau cleans, transforms, and prepares the dataset for visualization.
Dashboard Creation	Interactive maps, bar charts, pie charts, and

	filters are developed in Tableau.
Dashboard Publishing	The dashboard is published using Tableau Public.
Web Integration	The published dashboard is embedded into a Flask web application.
End User	Users interact with the dashboard through a web browser to analyze electricity consumption.

Working Process

- Collect electricity consumption dataset.
- Import and preprocess data in Tableau.
- Create interactive dashboards and visualizations.
- Publish the dashboard to Tableau Public.
- Embed the dashboard into a Flask web application.
- Users analyze electricity consumption through the web interface.

Benefits of the Architecture

- Interactive and user-friendly visualization.
- Modular and scalable architecture.
- Easy integration with Tableau Public and Flask.
- Supports future enhancements such as cloud deployment and predictive analytics.